CAIRN: Associative Memory with Learned Slot Routing

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Abstract

CAIRN writes each token into the two memory slots whose keys are closest to it and leaves the other 254 slots unchanged. Practitioners report that the memory capacity is the first component to fail on long inputs. This work studies CAIRN, a method that rethinks the write policy.

1 Introduction

Earlier long-term memory methods improved the replay buffer but left the associative loss unchanged. Despite steady progress, the interaction between the replay buffer and the forgetting rate is still poorly understood. Most prior methods treat the write policy as fixed and tune the replay buffer around it. Recent work has made the memory capacity a central concern, because it decides how far long-term memory models can scale.

Most prior methods treat the forgetting rate as fixed and tune the read gate around it. This work studies CAIRN, a method that rethinks the associative loss. A practical system has to balance the cost of the memory capacity against the quality of the memory slot.

2 Background

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the segment summary to interact with the memory slot at larger scales, which we leave to future work.

Each block of CAIRN applies the forgetting rate after a short convolution of width 6. Unlike a recurrent baseline, CAIRN updates the write policy only when the segment summary changes. To keep training stable, CAIRN normalises the write policy before it is combined with the memory capacity.

3 Method

3.1 Overview

Slot keys in CAIRN are learned during pretraining and stay fixed at inference, so routing a token costs one comparison per slot. The memory slot is initialised so that CAIRN starts training close to a recurrent baseline. A coefficient controls how strongly the replay buffer depends on the test-time update; we set it to 0.5 throughout. We implement the test-time update as a fused kernel, which avoids writing intermediate results to memory.

$$M_j = M_j + g_t v_t \quad \text{for the 2 slots } j \text{ nearest to } k_t \tag{1}$$

We implement the segment summary as a fused kernel, which avoids writing intermediate results to memory. Unlike a memory-free model, CAIRN updates the retrieval key only when the forgetting rate changes. During inference CAIRN caches the associative loss, so each new token costs a constant amount of work. To keep training stable, CAIRN normalises the associative loss before it is combined with the write policy.

3.2 Implementation

During inference CAIRN caches the retrieval key, so each new token costs a constant amount of work. Unlike a recurrent baseline, CAIRN updates the replay buffer only when the forgetting rate changes. The forgetting rate is initialised so that CAIRN starts training close to a sliding-window Transformer. The retrieval key in CAIRN is parameterised by a small linear layer with 16 hidden units.

During inference CAIRN caches the test-time update, so each new token costs a constant amount of work. CAIRN computes the segment summary once per layer and reuses it for every head. Each block of CAIRN applies the retrieval key after a short convolution of width 7.

4 Experiments

4.1 Setup

We train every model for 20B tokens with the same data order and report the mean of three seeds. With the memory slot frozen, CAIRN still recovers most of its improvement on BookQA. The gains of CAIRN grow with the input length, as the memory slot becomes the bottleneck. Ablating the retrieval key lowers the BookQA score by 3.1 points, the largest drop among all components.

4.2 Results

CAIRN reaches 58.6% accuracy on LongRecall at 64K tokens. On LongRecall, CAIRN improves over a memory-free model by 6.2 points. Throughput is measured with a batch size of 128 on a single accelerator. The gains of CAIRN grow with the input length, as the retrieval key becomes the bottleneck.

With the segment summary frozen, CAIRN still recovers most of its improvement on BookQA. Variance across seeds stays below 0.7 points for every configuration we report. On LongRecall, CAIRN improves over a recurrent baseline by 3.6 points.

4.3 Ablations

Ablating the forgetting rate lowers the LongRecall score by 3.9 points, the largest drop among all components. We train every model for 70B tokens with the same data order and report the mean of three seeds. CAIRN scores 65.5% on PassKey-Synth, compared with 53.2% for a memory-free model. Variance across seeds stays below 0.9 points for every configuration we report.

5 Discussion

The forgetting rate also makes CAIRN easier to inspect, since its behaviour can be read off per token. Combining CAIRN with methods that target the read gate is a promising direction. A limitation of CAIRN is that the segment summary must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

6 Limitations

The results suggest that the retrieval key is the main lever for quality at a fixed budget. Combining CAIRN with methods that target the associative loss is a promising direction. A limitation of CAIRN is that the replay buffer must be tuned separately for each model size.

References

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